National Science Foundation Request for Information on the Development of an Artificial Intelligence (AI) Action plan

Submission from iProov

13th March 2025

Key Message

iProov welcomes the opportunity to respond to the request for information for the AI Action Plan, issued by the National Science Foundation. The US has a clearly dominant position in AI by most relevant measures, and is well placed to reap economic and national security benefits from this hegemony, grabbing a major share of the new benefits as AI deployment fuels global economic growth. However, there are indicators that suggest there are challenges to the US sustaining this current position of strength. In this brief we highlight the threat posed to US dominance as a consequence of adoption and usage levels for AI in the US being below those of major rivals, such as China. Behind the data on relatively low rates of adoption is a high level of citizen mistrust in the use of AI by government and business. Moreover, malicious actors, including adversarial nations, are increasingly arming themselves with the same sophisticated AI tools for purposes of committing identity fraud to steal money, intellectual property, sensitive business information and threaten national security. Research shows that citizens with a high level of grievance against the government or business, for example as a consequence of a fear of fraud, have lower levels of adoption and use of AI tools. Helpfully, the same research also shows that by increasing levels of trust the negative impact of grievance can be outweighed, driving AI adoption and usage upwards.

As we explain, trust is an asset and serves to drive economic growth. Increasing the trustworthiness of AI tools and digitally enabled channels drives trust and therefore adoption and usage. The recommendations we make below are pragmatic and low-cost to government measures which would enhance trust in AI tools and digitally enabled channels by supporting citizens and organizations, including the government, in the battle against malicious use of AI tools.

Background on iProov

iProov provides mission-critical biometric face verification to protect governments and organizations from identity or imposter fraud. iProov uses advanced AI solutions to deliver the highest level of identity assurance by confirming that the individual is the right person, a real person, who is authenticating in real time. Our products and processes are regularly subject to external auditing and accreditation. iProov is a technology innovator and market leader and is established as a trusted service provider in major markets globally. Founded in the UK in 2012 and with a global presence including a growing team in the US, based out of Catonsville, Maryland, iProov is a research-led innovator with 24 registered patents. Not only is iProov a recognized world leader in identity verification, but we are also a highly successful supplier of services to important organizations in major markets. iProov customers include the U.S. Department of Homeland Security, the Australian Taxation Office, GovTech Singapore, the UK Home Office, the UK National Health Service (NHS) and others. We are a provider of identity verification for the US IRS, in partnership with ID.me, and our solution for strengthening border security is currently being trialled at major US airports, including Dulles and O'Hare. iProov's expertise is regularly recognised in analyst reports, and our class-leading role was most recently featured in Acuity Market Intelligence's 2024 Biometric Digital Identity Flagship Prism Report.

As the only biometric verification vendor with in-production data and intelligence through our cloud Security Operations Center (iSOC), we have been tracking the behaviors of threat actors over the last 13 years and have analyzed how their methodologies have evolved. Our commitment to advancing biometric security has led to rigorous independent testing by the US Department of Homeland Security's Science and Technology Directorate and cybersecurity leaders like Outflank, Jumpsec, and Kroll Redscan, validating our robust defense capabilities against emerging sophisticated attacks. We regularly advise key organizations and governments, such as ENISA and MITRE, helping raise awareness of real-world threats and establishing best practices for biometric security. We share intelligence publicly on how malicious actors are adopting the latest AI tools through public reports, such as our class-leading Threat Intelligence Report, as well as on measures to effectively protect organizations and citizens against malicious use of AI to create fraudulent identities.

The US Economic Opportunity Driving iProov

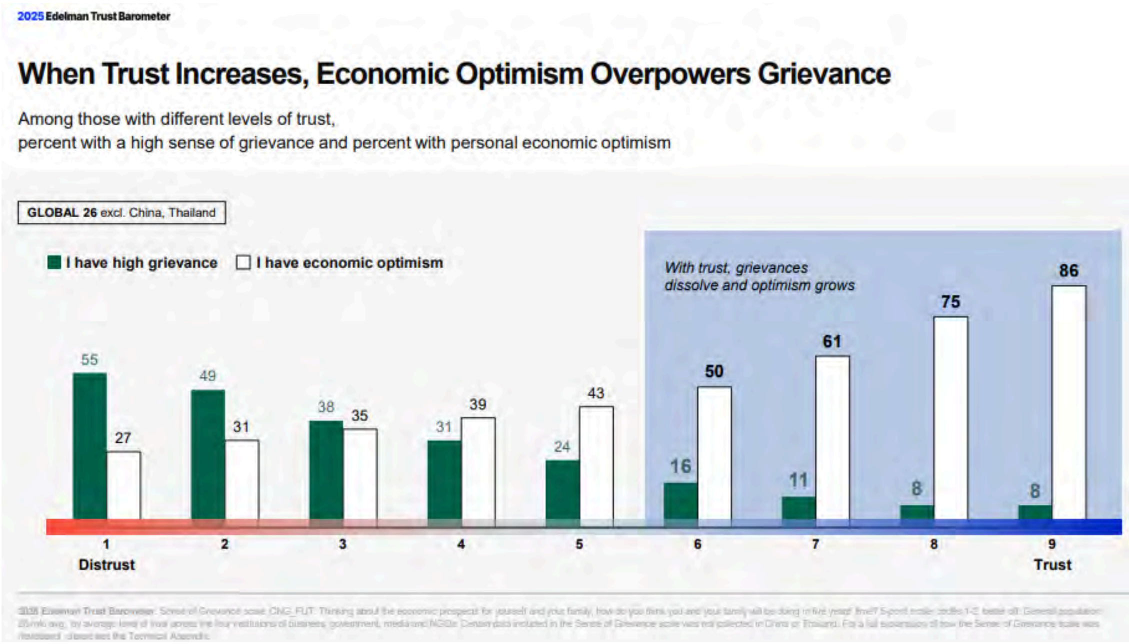
The US is the leading AI superpower, by most objective measures. According to the Global Vibrancy Tool 2024 from the Stanford Institute for Human-Centred AI, the US is the leading country for developing machine learning models (61 vs 13 from China in 2023), for AI investment (over $67bn in 2023, vs $7bn in China), more than times the number of data centers than the next largest country (Germany) and an increasing majority as a destination for elite AI talent (75% of the top-tier AI talent, up from 58% in 2019), based on research from think-tank Marco Polo. Moreover, the United States remains far and away the leading destination for the world's most elite AI talent (top 2%) and remains home to 60% of top AI institutions, according to the same tracker.[1]

US dominance is not unassailable over the long-term, in particular with China now leading the registration of AI patents, with more than 61% in 2023 (vs the US at 20%), and the US falling behind Singapore, South Korea, China and Denmark on the measure of registrations per capita. A more immediate threat to job creation and innovation in AI is posed by relatively poor adoption and usage rates, with SAS reporting that Generative AI use in surveyed US organizations (65%) lags behind levels in China (83%) and the UK (70%), and is only marginally ahead of Australia (63%).[2]

In order to address the challenge of adoption and usage, it is critical to understand what factors are driving the relatively low rates in the US. Edelman's 2024 Trust Barometer found that by nearly a three-to-one or more margin, respondents in the US (and other developed economies) reject the growing use of AI rather than embrace it. That contrasts to developing markets such as Saudi Arabia, India, China, Kenya, Nigeria and Thailand where acceptance is around

[1] Available at https://archivemacropolo.org/interactive/digital-projects/the-global-ai-talent-tracker/
[2] Available at https://www.sas.com/en_us/news/press-releases/2024/july/genai-research-study-global.html

two-to-one over resistance. It is noteworthy that Americans respondents were much more likely to cite reasons associated with grievances with the technology and accountability, such as potential harm to society (61%), privacy concerns (52%) and lack of adequate testing and evaluation (54%) than to a fear of negative consequences for their jobs.[3] Edelman reports that those with a higher sense of grievance or mistrust in Government or business have a lower level of trust in AI firms and in the use of AI by business. High levels of grievance with government and institutions is a global challenge, rather than one unique to the US, as reported in other global surveys.[4] However, Edelman's research reports that the dampening effects of high levels of grievance on AI adoption and use can be outweighed by the existence of high levels of trust.[5] The chart below illustrates Edelman's research findings.



When considering actions to prioritise, we would respectfully invite the NSF to consider this research (and similar) on the underlying causes of low rates of adoption and usage in the US, and support measures which would serve to counter. Specifically, we would propose that the NSF focus on supporting measures to enhance levels of trust in AI in the US.

The summarised research shows a significant positive relationship between trust and economic growth, which has long been understood by economists. As stated by Nobel Prize winning US economist Kenneth Arrow: "**virtually every commercial transaction has within itself an element of trust, certainly any transaction conducted over a period of time. It can be plausibly argued that much of the economic backwardness in the world can be explained by the lack of mutual confidence.**"[6] The relationship is also supported by empirical research

[3] Available at https://www.edelman.com/trust/2024/trust-barometer
[4] See, in particular, *OECD Survey on Drivers of Trust in Public Institutions - 2024 Result*. Available at https://www.oecd.org/en/publications/oecd-survey-on-drivers-of-trust-in-public-institutions-2024-results_9a2055b-en.html
[5] Available at https://www.edelman.com/trust/2025/trust-barometer
[6] Arrow, K. J. (1972). *Gift and Exchanges*. Philosophy & Public Affairs, 1(4), 343-362. Retrieved from https://www.jstor.org/stable/2265097

showing that economies with higher levels of trust have higher levels of per-capita Gross Domestic Product (GDP).[7] Figure 1, taken from Zakharov et al[8], illustrates the relationship.

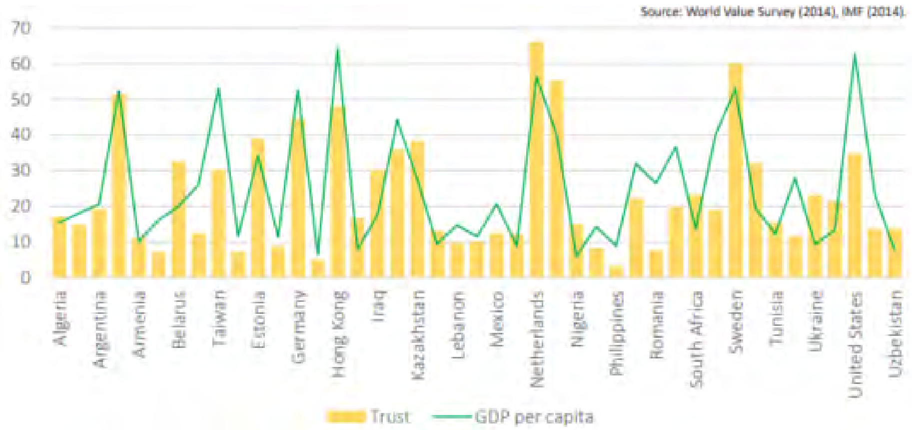


Figure 1. The correlations between trust and GDP per capita

The relationship is also causative. Higher trust levels lead to more efficient resource use as a consequence of reduced free riding, lower transactions costs and lower contract enforcement costs. In the context of digital and AI enabled channels, the typically transient nature of commercial relationships and absence of a physical engagement means that such compliance costs are greater than is the case in physical channels. By increasing trust, valuable resources can therefore be diverted to activities which create more wealth and income.[9]

By raising levels of trust in digital and AI enabled channels, reduced barriers to efficient transacting between organizations and between individuals generates improvements in Total Factor Productivity (TFP).[10] Coyle and Lu (illustrated in Figure 2) model the positive impact of trust on productivity (TFP), which feeds through to increased per capita GDP in their multi-country study.[11] Similar findings are presented in other studies.[12][13]

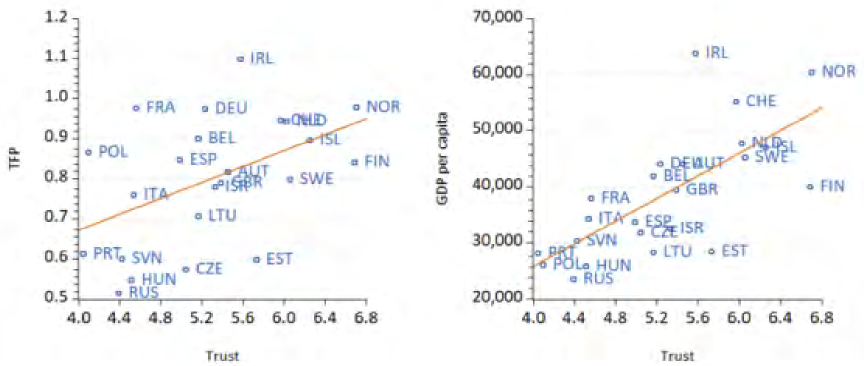


Figure 2 - Trust levels driving TFP and GDP per capita

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[7] Per capita GDP is generally accepted as the best metric for measuring economic growth.
[8] Dmytro Zakharov, Svitlana Bezruchuk, Viktoriia Poplavska, Svitlana Laichuk and Hanna Khomenko (2020). *The ability of trust to influence GDP per capita*. Problems and Perspectives in Management, 18(1), 302-314. doi:10.21511/ppm.18(1).2020.26
[9] Dasgupta, P, *A Matter of Trust: Social Capital and Economic Development*, May 2009, available at https://www.econ.cam.ac.uk/people-files/emeritus/pd10000/publications/09/abcde09.pdf
[10] Total Factor Productivity (TFP) is a measure of efficiency, derived by comparing total outputs relative to the total inputs used to produce those outputs.
[11] Coyle, D & Lu, S (2020), *Trust and Productivity Growth - An Empirical Analysis*, available at https://www.bennettinstitute.cam.ac.uk/wp-content/uploads/2020/12/Trust_and_Productivity_Coyle_Lu.pdf
[12] Knack, S., & Keefer, P. (1997), *Does social capital have an economic pay-off? A cross-country investigation*. Quarterly Journal of Economics, 112(4), 1251-1288. Retrieved from https:// www.jstor.org/stable/2951271
[13] Whiteley, P. F. (2000), *Economic growth and social capital*, Political Studies, 48(3), 443-466.

Critically for the purposes of retaining US AI dominance, organizations in markets with higher levels of trust also benefit from a lower cost of equity, which may reduce cost barriers to investment and innovation (and job creation).[14] Firms operating in markets with higher trust levels may be more profitable as a consequence.[15] To illustrate the magnitude of the potential impact of improved trust on growth, the World Economic Forum (WEF) estimates that (in 2019 values) **"a 5% point increase in digital trust results in an average increase in GDP per capita of $3,000".**[16] McKinsey and others have made similarly upbeat estimates of the impact of AI on the US economy.[17] Microsoft estimates that Generative AI could contribute $3.8 trillion to the US economy by 2038.[18] McKinsey suggests AI innovation could unlock between $11-$17 trillion in increased total global economic value for business.[19] However, benefits will not be evenly distributed, with the greatest gains going to those countries where adoption and usage is highest. While this would suggest that the US economy will continue to benefit significantly from AI, its hegemony is at risk due to lower rates of adoption and use than many rivals, with the gap to China increasing year-on-year.

Measures to Enhance Trust in AI will Drive Economic Growth, Innovation and Job Creation, as well as National Security

The AI Action Plan provides a critical opportunity to establish a pro-innovation and pro-growth policy environment for the purposes of underpinning current US dominance in AI and securing this lead into the future. Economic research shows that for the US to secure the economic growth potential of AI it is the trust of users (US citizens and organizations) which is crucial to driving adoption and usage to levels which will incentivise innovation and reward investment. However, trust itself is an outcome with trustworthiness a necessary precondition. Those very same AI tools which can fuel the sorts of economic growth the US has seen in recent decades, can also be used by malicious actors to render AI tools and AI-enabled digital channels untrustworthy. The rapid growth of AI-based technology has introduced new challenges for remote identity systems, undermining the trustworthiness of digital economy channels. Innovative and easily accessible tools have allowed threat actors to become more sophisticated overnight, powering an increasing number of threat vectors due to new methodologies. The growth of new attack vectors over the last 24 months has heavily impacted organizations. The Federal Trade Commission's Consumer Sentinel Network documented a 45% increase in identity theft incidents in early 2024, with aggregate fraud losses exceeding $10.2 billion.[20] The second-highest reported loss amount came from imposter

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[14] Gupta A, Raman K & Shang C (2018),*Social capital and the cost of equity*, Journal of Banking & Finance, Volume 87, 2018, Pages 102-117, ISSN 0378-4266, https://doi.org/10.1016/j.jbankfin.2017.10.002. (https://www.sciencedirect.com/science/article/pii/S0378426617302479)

[15] Hilary, G., & Huang, S. (2016). *Trust and Contracting,* (INSEAD Working Paper No.2015/42/ACC). Retrieved from https://papers.ssrn. com/sol3/papers.cfm?abstract_ id=2604974

[16] *Digital trust: How to unleash the trillion-dollar opportunity for our global economy*, World Economic Forum, August 2022. Available at https://www.weforum.org/agenda/2022/08/digital-trust-how-to-unleash-the-trillion-dollar-opportunity-for-our-global-economy/

[17] For example, see https://www.thirdway.org/report/ai-and-the-us-economy-optimism-pessimism-or-realism

[18] Available at https://blogs.microsoft.com/on-the-issues/2024/11/21/the-3-8-trillion-opportunity-unlocking-the-economic-potential-of-the-us-generative-ai-ecosystem/

[19] Chui, Michael, et al. "The economic potential of generative AI: The next productivity frontier." *McKinsey Digital*, McKinsey & Company, 14 Jun 2023, https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-economic-potential-of-generative-ai-the-next-productivity-frontier#business-value.

[20] As Nationwide Fraud Losses Top $10 Billion in 2023, FTC Steps Up Efforts to Protect the Public | Federal Trade Commission

scams, with nearly $2.7 billion in reported losses, indicating a significant upward trajectory in financial impact. There is a technological arms race at play, with responsible organizations (including the US Government) investing in measures to protect their systems from malicious actors. However, even against this background of action and high profile support for tighter controls on identity fraud from agencies such as FinCen, fraud levels continue to grow.[21] In more recent reporting, the Federal Trade Commission claims that consumer losses in 2024 due to fraud have increased by 25% in the preceding 12 months to $12.5bn. Losses due to imposter scams remains the second-highest category reported, at $2.95bn, an increase against the losses claimed in the preceding report (referred to above).[22]

Malicious actors using these tools also pose a real and present threat to US national security. Nation states which are adversarial to the US are increasingly making use of these same tools to attack US interests, put US security at risk and defraud US taxpayers. Recent reports of North Korean cyber attacks on the US, whereby hackers stole the identities of US citizens and employed sophisticated face-swapping methods to secure jobs with US organizations for the purposes of stealing sensitive information, technical intellectual property and money from US companies and taxpayers, serve to illustrate a growing national security threat posed by malicious use of AI tools for the purposes of identity or imposter fraud. [23]

As the FTC correctly notes, "*scammers' tactics are constantly evolving*". Synthetic identity fraud (SIF) is the fastest-growing type of fraud, combining legitimate data (such as valid Social Security numbers often stolen from children, elderly, or deceased individuals) with fabricated personal information to create convincing false identities. What makes SIF especially challenging to combat is its ability to evade traditional fraud detection systems. Unlike conventional identity theft, where systems can flag stolen information based on reports from real victims, SIF uses AI to create entirely new identities that incorporate both real and fake elements. Without a real victim to raise the alarm, and with some components of the identity being legitimate, traditional detection methods often fail to recognize these synthetic fraud patterns.

Democratization and ease of use of AI tools, such as applications for native camera spoofing and face swaps, has fuelled the growth of online Attacks-as-a-Service communities. Malicious actors look to bypass security methods by combining these AI tools. We track 127 face swap tools, 10 mobile device emulators and 91 virtual camera tools. At a basic level, this means that there are 115,570 potential attack combinations using readily available (and often free to use) AI tools. As new tools and updates are introduced, combinations are constantly increasing.

A 2025 deepfake consumer research report by iProov paints a picture of a society largely unprepared for the challenges posed by deepfake technology, with significant gaps in

[21] See, for example https://www.fincen.gov/news/news-releases/fincen-issues-analysis-identity-related-suspicious-activity
[22] Available at https://www.ftc.gov/news-events/news/press-releases/2025/03/new-ftc-data-show-big-jum-reported-losses-fraud-125-billion-2024
[23] See for example, https://www.justice.gov/archives/opa/pr/fourteen-north-korean-nationals-indicted-carrying-out-multi-year-fraudulent-information

awareness, detection abilities, and response mechanisms.[24] Only 0.1% of participants (including those from the US) could identify all synthetic media examples correctly, and there was a particularly low success rate (9%) for video deepfake detection. Against this background of increasing complexity and human vulnerability, it is important to take a multi-faceted and evidence-based approach to safeguarding trust in AI and digital channels.

Recommendations for Priority Actions to Secure Trust in AI

An AI Action Plan which prioritizes measures to enhance capabilities and trust in the use of AI-enabled channels would fuel US AI innovation and underpin continued leadership. We invite the NSF to consider four recommendations which we believe would enhance trust in AI and, as illustrated by economic research, propel innovation and economic growth.

1. Facilitate Research and Development in AI-Resistant Identity Verification

AI-based attack tools are becoming democratized and more sophisticated. Many of the current identity verification and KYC methods deployed by organizations, including banks, law enforcement and government, are not sufficient to keep pace with these evolving threats. We would encourage the NSF to facilitate and sponsor industry-led research and development of novel identity verification methods specifically designed to be robust against AI-generated deepfakes, synthetic identities, and other advanced spoofing techniques. This should include research into biometric technologies, liveness detection, and cryptographic solutions.

2. Development of Standardized Evaluation Metrics and Testing Protocols for AI-Based Identity Verification Systems

A significant challenge is posed by inflated vendor claims and the consequent difficulty organizations face in procuring appropriate security technologies. NIST should be tasked with extending its world-leading efforts to establish standardized evaluation metrics and testing protocols to assess the effectiveness of AI-based identity verification systems against sophisticated AI-driven attacks, including those based on synthetic media and digital injection attacks.

3. Strengthening Public-Private Partnerships for Threat Intelligence Sharing and Collaborative Defense

Our work has served to show the importance of real-time threat intelligence and the need for continuous monitoring and adaptation. We would encourage the NSF to work collaboratively across stakeholders to foster stronger public-private partnerships for the purpose of facilitating the sharing of threat intelligence related to AI-based identity fraud. This should include the establishment of a central repository for reporting and analyzing attacks, as well as mechanisms for rapid dissemination of threat information to relevant stakeholders.

4. Focus on Education and Training to Address the Knowledge Gap in AI Security

Our recent Threat Intelligence Report identifies a fundamental knowledge gap regarding understanding and procuring appropriate remote verification technologies. We would therefore recommend that the NSF partner with relevant agencies and the private sector to support the

[24] https://www.iproov.com/press/study-reveals-deepfake-blindspot-detect-ai-generated-content

development and rolling out of education and training programs to raise awareness of AI-driven threats and best practices for securing identity verification systems. This should target both technical professionals and non-technical decision-makers within organizations. It could be operated on a verticals-basis, for example with FinCen overseeing the approach for regulated financial services providers, Homeland Security responsible for the approach to border control, national security and law enforcement, as well as horizontally under the leadership of the FTC.

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